

Homework

Prof. Dr. Paula Cocoma

Portfolio Management - Fall 2026

This assignment focuses on the time-series analysis of stock returns and portfolio construction. It counts for 40 points of your final grade and the assignment is graded out of 100%. Please work in groups of one to three students.

You may use LLMs during the assignment, however, beware that they can (and tend to) make mistakes. You, or at least a member of your group, are requested to attend the defense session (scheduled for October 13th). In case that a group fails to defend their homework, either by not being present or not being able to answer the question(s) asked, the *whole* team will incur a grade penalty of 20%.

Please hand in the assignment as a pdf file before the deadline in Canvas. Only the pdf file will be graded, but you are free to use Excel, Python, R or alike for solving it and submit the associated files as additional material on Canvas. The TA for this assignment is Maria Chiara Leone (m.leone@fs.de). Late assignments **will not** be accepted.

If you have any clarification questions, please check the link: [clarify](#), please

There are five datasets for this assignment:

1. The first file, HW_Factors.csv, contains monthly returns of the four empirical factors (Market, SMB, HML and MOM) and the 1-month Treasury bill rates from Jan 1927 to May 2025. The market returns are reported in excess of the 1-month Treasury bill rates. All returns in this file are in percentage.
2. The second file, HW_Hedge Fund.csv, contains the monthly rate of return of eight Hedge Fund Indexes (from HFRI).
3. The third data file, HW_World.csv, contains the monthly equity returns of 20 developed countries. The equity returns in this dataset have all been converted to U.S. dollar-based returns. In other words, you need not worry about exchange rate fluctuations. All returns in this file are in percentage.
4. The fourth file, HW_Prices.csv, contains historical stock price data from June 2004 to December 2022 of 30 stocks that belong to the Dow Jones

Industrial Average, including their name, ticker, and high group industry classification. It also includes the number of shares outstanding as of Jan 2004 and Dec 2024. The last column of the file contains the DJIA index over the same period.

5. The fifth file, HW ETFs.csv, contains the time series prices of 16 different ETFs, names in the first row. Note that the length of data availability is not homogeneous, as to preserve realism.

One note of caution: some returns are reported in percent while others are in decimal points. Be careful when you subtract one number from another. For example, you might want to subtract the risk-free rate from a hedge fund's return to get its excess return.

Part 1: Hedge Funds (20%)

Use files (HW_Hedge Fund.csv; HW_Factors.csv).

The main objective is to get you familiarized with the hedge fund industry: the overall performance of the industry, as well as the various styles of hedge funds.

1. CAPM (5%)

Report the CAPM beta, alpha, t-stats, and R-squared of each index. How do those numbers for the aggregate Hedge Fund Index compare to your knowledge of the Mutual Fund industry? How good is CAPM at explaining the different hedge funds' returns? (Hint: do a scatter plot)

2. Four-Factor Model(5%)

Repeat the same analysis under a four-factor model (regression and scatter plot). Comment on the effect of the four factors on the aggregate index and how the different Hedge Fund indices load on the different factors. Is the four-factor model better than CAPM at explaining the different hedge funds' returns?

3. Time-Varying Beta (5%)

The market exposure for hedge funds might be time-varying. For example, hedge funds might have high market exposure during down markets and low exposure during up markets. Use the market return to identify up and down markets, calculate the CAPM beta in the two cases, and compare them. What does this mean in terms of the attractiveness of hedge funds?

4. Autocorrelation (5%)

Hedge funds' returns might display serial autocorrelation. Run an AR(1) for each of the Hedge Fund indices and report the significance of the autocorrelation. What can you conclude from such evidence?

Part 2: International Momentum (25%)

Use files (HW_World.csv; HW_Factors.csv).

The main objective is to investigate whether or not there are momentum profits in the equity markets across the 20 developed countries. Let's use the following strategy to form our momentum portfolio:

1. In month t , sort the indices according to the 11-month cumulative return from month $t - 12$ to month $t - 2$ (month $t - 1$ is skipped).
2. Pick the three best and three worst performing countries according to the sort.
3. Form a long-only portfolio: 80% of your portfolio comprises the three best-performing countries equally weighted (weights=4/15), 0% of your portfolio is assigned to the three worst-performing countries. The remaining 14 countries account for the other 20% of your portfolio, again equally weighted (weights=1/70).
4. Hold this long/short portfolio for month t . Then in month $t+1$, resort and rebalance using the same momentum strategy. Assuming no transaction costs.

How does this strategy perform? (provide a cumulative return plot, average annualized return, volatility, maximum drawdown and Sharpe ratio) Can the four-factor model explain the international momentum profit? Summarize your findings and compare to the factors.

Part 3: Portfolio Replication (25%)

Use files (HW_Prices.csv).

The main objective is to get you familiarized with the data of the Dow Jones Industrial Average Index, especially compared to the index performance, which is investible through ETFs on the index. Use the data set HW_Prices.csv and assume costless monthly rebalance unless otherwise specified. The risk-free rate is zero.

1. Different Portfolios (10%)

Compute the following portfolios and report their weights at inception.

- i The market-cap weighted portfolio, using the share outstanding and prices as of Jan 2024.
- ii The market-cap weighted portfolio, using the share outstanding as of Jan 2024 but the prices in the middle of the sample (June 2014) .
- iii The equal-weighted portfolio among the 15 stocks with names higher in the alphabet, zero weights in the rest.

- iv The equal-weighted portfolio among the 15 stocks with names lower in the alphabet, zero weights in the rest.
- v The tangent portfolio, using half the sample (until June 2014) to estimate expected returns and the variance-covariance matrix.
- vi The minimum-variance portfolio, using half the sample (until June 2014) to estimate the variance-covariance matrix.
- vii The naive risk-parity portfolio (with inverse-volatility weights), using half the sample to estimate volatilities.
- viii The most diversified portfolio, using half the sample (until June 2014) for estimates.
- ix The equal-budget group portfolio, where each of the five groups receives 1/5 of the total weight and stocks within each group are equally weighted. Use the major group variable provided in the data set.
- x The equal-risk-contribution portfolio, where each of the 30 stocks groups contributes 1/30 of total variance. Use half the sample (until June 2014) to estimate volatility and correlations.

2. Compare Portfolios (5%)

Which of the previous portfolios are comparable, and which ones aren't?

3. Performance Measures (10%)

Using your constructed portfolios and the DJIA index, compute the mean return, volatility, skewness, minimum monthly return, and Sharpe Ratio, assuming a risk-free rate of zero and using the full sample (from Jan 2004 to Dec 2024). Furthermore, compute the investor's utility assuming a mean-variance utility function with a risk aversion of 5. Plot the performance of these portfolios. In which of them would you invest on January 1st 2004?

Part 4: Retirement Management (15%)

Use files (HW ETF.csv).

In this exercises you are a financial advisor and your client is a US employee saving for retirement in their tax-free 401K pension account. To structure her portfolio you use ETFs due to their liquidity, low transactions costs and transparency. Use the data set called HW ETFs.csv. You will construct and compare the historical performance of the following portfolios, assuming costless monthly rebalancing:

- i The baseline 60/40 portfolio of 60% equity and 40% bonds that invest all the equity in the S&P 500 (SPY) and all the bonds in the long-term treasuries (TLT)

- ii A portfolio that keeps the 40% bonds allocation in the long-term treasuries but allocates the 60% equity as follows: 50% US equity, approximated by the Russell 2000, 25% of allocated to Europe, approximated by the STOXX 50 Europe , and 25% to emerging market equity.
- iii A portfolio that keeps the 60% equity allocation purely in the S&P 500 but allocates the 40% bond by equally weighting each of the three Fixed Income ETFs provided (AGG, JNK and LQD), i.e., allocate 33.3% of the bond portion to each of them.
- iv A portfolio that keeps the 60/40 policy of 60% equity and 40% bonds but inside the equity portion assigns equal weight to each of the four equity ETFs provided and inside the bond, portion assigns equal weights to each of the four Fixed Income ETFs provided.
- v A portfolio that keeps the 40% bonds allocation in the long-term treasuries but allocates the 60% equity by equally weighting each of the four equity factors ETFs (Momentum, Size, Value and min. volatility), i.e., allocate 25% of the equity portion to each of them.
- vi A portfolio that keeps the 60% equity allocation purely in the S&P 500 but allocates the 40% by equal weighting each of the alternatives ETFs (REIT, Hedge Funds, Infrastructure and Private Equity).
- vii A portfolio that breaks the 60/40 policy by equal weighting all sixteen ETFs provided

Calculate the annualized returns, volatility as well as the maximum draw-down return and Sharpe ratio with a risk-free rate of 1%. Plot and compare the performance and discuss the reasons behind the differences with the baseline portfolio.

Part 5: Windfall Investment (15%)

Use files (HW_Factors.csv).

Assume you unexpectedly receive a windfall of €100,000 (for example, an inheritance). Evaluate the following investment strategies.

1. Policy Portfolios (10%)

You decide to invest the entire amount immediately and hold the portfolio for the next 30 years without making any additional contributions, withdrawals or rebalancing. Because the outcome depends on the date at which the investment begins, assume that the windfall is equally likely to occur at the end of any month in the data set (Jan 1927 to April 1996), provided that at least 30 years of subsequent data are available.

Consider a family of traditional stock–bond policy portfolios using the Market Factor (from Fama-French) and the risk-free rate. Construct eleven portfolios, beginning with a 100% bond portfolio and ending with a 100% equity portfolio, increasing the equity allocation in increments of 10 percentage points (i.e., 0/100, 10/90, 20/80, ..., 90/10, 100/0).

For each portfolio, calculate the average (over the 832 months between Jan 1927 and Apr 1996) cumulative balance of your €100,000 after 30 years, and plot it as a function of the time when you received the windfall. Calculate as well the average annualized return, average annualized volatility, and average annualized Sharpe ratio, assuming a constant risk-free rate of 1%, and provide plots for these three metrics over the sample period (Caution: these are not the same as for a portfolio with monthly rebalance, since the weights are drifting with the cumulation). Discuss the trade-offs between risk and return across the different policy portfolios. Which one would you choose if you received such a windfall today?

2. Market Timing (5%)

Suppose you believe you can successfully time the market. Compare the following investment strategies:

1. A passive strategy that remains invested 100% in the equity market (Market Factor) throughout the entire sample (starting Jan 1927 and holding until April 2026).
2. A market-timing strategy that remains invested 100% in bonds (Risk Free Rate), except during the historically best month of the equity market, when it switches to 100% equities before returning to bonds the following month. Assume there are no transaction costs.

Repeat the analysis assuming you correctly identify, the best two months, the best three months, the best four months, etc. of the entire sample.

How many months would an investor have to predict perfectly in order to outperform the passive buy-and-hold equity strategy?